INCH-POUND

MIL-M-46861B (MI)
13 February 1991
SUPERSEDING
MIL-M-46861A (MI)
7 March 1980

MILITARY SPECIFICATION

MOLDING MATERIAL, GLASS MAT, EPOXY COATED

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of epoxy coated glass mat molding materials.

1.2 Classification. The material shall be of the following types, as specified (see 6.2):

Type I	Mat, epoxy coated on the unidirectional roving side
Type II	Mat, epoxy coated on the chopped fiber side

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN; AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898- 5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or letter.

AMSC N/A FSC 9330

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-R-9300	-	Resin, Epoxy, Low-Pressure Laminating
MIL-M-43248	-	Mats, Reinforcing, Glass Fiber
MIL-R-60346	-	Roving Glass, Fibrous (for Prepreg Tape and Roving, Filament Winding, and Pultrusion Applications)

STANDARDS

MILITARY

MIL-STD-1190	-	Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg., 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 613	Resin content of Carbon and Graphite Prepreg by Solvent Extraction, Standard Test Method for
ASTM D 638	Tensile Properties of Plastics, Standard Test Method for
ASTM D 695	Compression Properties of Rigid Plastics, Standard Test Method for
ASTM D 790	Flexural Properties of Plastics and Electrical Insulating Materials, Standard Test Method for

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ASTM D 792 — Specific Gravity (Relative Density) and Density of Plastics by Displacement, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material.

3.2.1 Type I composition. The type I composition shall consist of the reinforcement coated with a hot melt epoxy resin system on the unidirectional roving side (see 4.8.1).

3.2.2 Type II composition. The type II composition shall consist of the reinforcement coated with a hot melt epoxy resin system on the chopped fiber side (see 4.8.1).

3.2.3 Reinforcement.

3.2.3.1 Construction. The reinforcement shall meet the requirements of MIL-M-43248, type V, grade A, and table I of this document (see 4.8.3).

3.2.3.2 Component distribution. The roving shall be evenly distributed across the width of the material at an average concentration of 5 ± 1 roving per inch; a four inch width of material shall contain 20 ± 2 rovings. The chopped strand fiber shall be equally distributed over the entire surface. Surfacing mat (table II) shall cover the entire area of roving side and may be overlapped to accommodate finish rovmat widths 50 inches or greater (see 4.8.12).

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TABLE I. Reinforcement construction.

Component	Definition
Unidirectional roving - continuous	8-end, K-37 fiber, per MIL-R-60346, type I,class 1
Chopped strand	G-(135-140) fiber, per MIL-R-60346, type I, class 1 - 1 to 4 inches, random length
Surfacing mat	0.010 with compatible finish
Glass sizing - all reinforcement	Chrome silane or silane finish, 3% by weight maximum
Mechanical binding	Mechanical needling, approximately 180/square inch, over entire surface area through all reinforcing components
Width	As specified ± 0.25 inches (see 6.2)
Selvage	None - clean cut with no loose roving
Length	50 ± 10 linear yards. One splice per roll allowed. Flag each splice at roll end.

3.2.3.3 Roving distribution. The material shall be such that 80 percent of the construction is free of roving crossover. The gap between roving shall be not greater than 0.150 inch. A fine thread of single glass fiber, K-37 , may be interlaced at 90º to the unidirectional roving at intervals of approximately 0.5 inch to control roving spacing (see 4.8.13).

3.2.3.4 Weight. The weight of the bare material shall be 2.50 ± 0.25 ounces per square foot (see 4.8.14).

3.2.4 Resin system. The resin system shall consist of epoxy resin conforming to MIL-R-9300, type I, grade 0, form A, modified with a suitable elastomer (see 4.8.2).

3.2.5 Storage life. The materials as furnished by the manufacturer shall be capable of meeting the requirements specified herein after storage in the original unopened containers at a temperature not greater than 4º Celsius (C) for not less than 90 days (see 4.8.4).

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3.3 Performance characteristics.

3.3.1 Physical properties. The properties of the materials shall be as specified in table II.

TABLE II. Physical properties.

Characteristics	Average		Individual		Test paragraph
	Min.	Max.	Min.	Max.	
Tensile strength, ultimate psi x 10^3	31	--	25	--	4.8.5
Flexural strength, ultimate psi x 10^3	45	--	40	--	4.8.6
Flexural modulus, psi 10^6	1.7	2.7	1.5	--	4.8.6
Compression strength, ultimate psi x 10^3	26	--	23	--	4.8.7
Compression modulus, psi x 10^6	2.1	--	1.5	--	4.8.7
Volatiles content, percent	--	0.5	00	0.6	4.8.8
Resin content, percent	42	52	40	55	4.8.9
Gel time, minutes	2.5	5.0	2.0	5.5	4.8.10
Specific gravity	1.63	--	1.60	--	4.8.11

3.4 Environmental conditions.

3.4.1 Physical properties at high and low temperatures. The properties of the materials shall be as specified in table III. The requirements shall be met when tested at the specified temperature after a 0.5 hour soak (see 4.8.6).

3.5 Workmanship. The workmanship shall be such as to ensure a product which is uniform and in accordance with this specification. All the materials shall be free of dirt, foreign matter, or other contaminants (see 4.9).

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TABLE III. Physical properties at high and low temperatures.

Characteristics	Average Minimum		Individual Minimum		Test Paragraph
	at 68.3ºC	at -53.9ºC	at 68.3ºC	at 53.9ºC	
Flexural strength ultimate, psi x 10^3	30	58	22	50	4.8.6
Flexural modulus, psi x 10^6	1.1	2.1	0.9	1.8	4.8.6

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified herein, the following conditions shall be used:

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a.	Temperature	Ambient room 16 to 32ºC
b.	Altitude	Facility ground
c.	Humidity	Facility ambient up to 95 percent relative humidity

4.4 First article inspection. The contractor shall submit a first article sample for inspection as specified. The first article sample shall consist of not less than 5 yards of full width type I or type II material manufactured under the same conditions as those proposed for subsequent production.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table IV.

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TABLE IV. Quality conformance inspection.

Inspection	Requirement paragraph	Test paragraph
Composition Type I	3.2.1	4.8.1
Composition Type II	3.2.2	4.8.1
Reinforcement construction	3.2.3.1	4.8.3
Component distribution	3.2.3.2	4.8.12
Roving distribution	3.2.3.3	4.8.13
Weight	3.2.3.4	4.8.14
Resin system	3.2.4	4.8.2
Storage life	3.2.5	4.8.4
Tensile strength	3.3.1	4.8.5
Compression strength and modulus	3.3.1	4.8.7
Flexural strength, ult. x 10^3	3.3.1	4.8.6
Flexural mod. x 10^6	3.3.1	4.8.6
Resin content	3.3.1	4.8.9
Volatiles content	3.3.1	4.8.8
Gel time	3.3.1	4.8.10
Specific gravity	3.3.1	4.8.11
Physical properties at high and low temperatures	3.4.1	4.8.6
Workmanship	3.5	4.9

4.5.1 Sampling. Unless otherwise indicated in the contract or order (see 6.2), sampling shall consist of a random sample selected from each production lot in accordance with table V.

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TABLE V. Sample plan for quality conformance.

Lot Size	Sample Size
2 - 15	2
16 - 25	3
26 - 90	5
91 - 150	8
151 - 280	13
281 - 500	20
501 - 1200	32
1201 - 3200	50
3201 - 10,000	80
10,001 - 35,000	125
35,001 - 150,000	200
In all cases: Acceptance number is ZERO Rejection number is ONE	

4.5.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process by the same manufacturer in accordance with this specification and submitted for inspection at one time.

4.6 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods paragraphs (see 4.8) to verify conformance with requirements in section 3.

4.7 Test specimen preparation. Unless otherwise specified herein, specimens shall be prepared as follows to verify conformance with requirements in section 3.

a. A 12 x 12 inch flat panel shall be molded from the material as follows:

1. Four plies, 12 x 12 inches, shall be cut form the sample.

2. The panel shall be layed up so that the unidirectional rovings in the first three plies are parallel and in the fourth ply (outside) at 90º.

b. The panel shall be cured by placing the lay-up in a 12 inch mold preheated from 65 to 74ºC, and applying sufficient pressure to close the mold to stops which will control the panel thickness to 0.117 ± 0.005 inch. The panel shall be cured for 25 + 5 or
- 0.0 minutes.

c. The cured panel shall be removed from the mold and allowed to cool to ambient room temperature.

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d. Mechanical test specimens shall be cut from the panel so that the three parallel plies are parallel to the specimen length.

4.8 Test methods.

4.8.1 Composition, type I and II. The composition shall be determined visually to verify conformance to 3.2.1 or 3.2.2, as applicable.

4.8.2 Resin system. The resin system shall be certified by the supplier to verify conformance to 3.2.4.

4.8.3 Reinforcement construction. The reinforcement construction shall be certified by the supplier to verify conformance to 3.2.3.1.

4.8.4 Storage life. A sufficient amount of materials in the original unopened containers shall be stored under the conditions specified in 3.2.5. At the end of the storage period, the material shall meet the requirements specified herein. The supplier's certification of compliance to the storage life requirements (see 3.2.5 and 6.2) may be accepted during the storage period.

4.8.5 Tensile ultimate. The ultimate tensile strength shall be determined in accordance with ASTM D 638 and shall conform to the requirements specified in 3.3.1.

4.8.6 Flexural ultimate and modulus. The ultimate flexural strength and modulus shall be determined in accordance with ASTM D 790. The specimen shall be oriented to load the 90º ply in compression. The material shall conform to the requirements specified in 3.3.1 or 3.4.1, as applicable.

4.8.7 Compression ultimate and modulus. The ultimate compression strength and modulus shall be determined in accordance with ASTM D 695. A minimum of five specimens shall be tested. The material shall conform to the requirements specified in 3.3.1.

4.8.7 Volatiles content. The volatiles content shall be determined as follows:

a. Specimen size shall be 4 x 4 inches selected from random area in the sample.

b. the specimens shall be weighed to the nearest 0.01 gram.

c. Place the specimens on a rack in a circulating air oven preheated from 149 to 165ºC.

d. Allow the specimens to remain in the oven for 15 ± 1 minute.

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e. Remove the specimens and place in a desiccator until cooled to ambient room temperature.

f. Reweigh the specimens to the nearest 0.01 gram.

g. Calculate the volatile content using the formula:

$$C_V\ (\%) = \frac{W_O - W_1}{W_O} \times 100$$

Where:

C_V (%) = Volatile content percent

W_O = Original specimen weight

W_1 = Devolatilized specimen weight

h. Report the individual and average of five specimens. The volatiles content shall be as specified in 3.3.1.

4.8.9 Resin content. The resin content shall be determined in accordance with ASTM D 613. Specimens shall be those devolatilized specimens used in the volatiles determination of 4.8.8. The resin content shall be as specified in 3.3.1.

4.8.10 Gel time. The gel time shall be determined as follows:

a. Specimens approximately 2 x 2 inches shall be selected from random areas in the sample.

b. The specimen shall be placed resin side down on a hot plate preheated from 149 to 165ºC.

c. After three to five seconds, remove the sample, leaving a small amount of resin on the surface of the hot plate.

d. Start the stop watch.

e. Probe the resin until it will no longer form strings when the probe is removed.

f. Stop the watch and record the gel time.

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g. Report the individual and average of five tests. The gel time shall meet the requirements specified in 3.3.1.

4.8.11 Specific gravity. The specific gravity shall be determined in accordance with ASTM D 792. Specimens shall be cut from the end of the specimens used to determine flexural strength in 4.8.6. A minimum of five specimens shall be tested. The specific gravity shall be as specified in 3.3.1.

4.8.12 Component distribution. The roving concentration shall be determined on
the bare mat specimens remaining from the resin content determination of 4.8.9. The rovings contained in the three 4 x 4 inch specimens shall be counted and recorded, and shall conform to the requirements specified in 3.2.3.2. The chopped strand distribution and surface mat shall be determined visually, and shall conform to 3.2.3.2.

4.8.13 Roving distribution. The roving distribution shall be determined visually, and shall conform to 3.2.3.3.

4.8.14 Weight. The mat weight shall be determined as follows:

a. Using the bare mat specimens from the resin content determination of 4.8.9, weigh each to the nearest 0.01 gram.

b. Calculate the weight using the following formula:

$$W_m = 0.317\ W_f$$

Where:

W_m = Mat weight - ounces per square foot

W_f = Bare mat weight - 4 x 4 inches specimen

0.317 = Constant for converting grams per specimen to ounces per square foot

c. The weight shall be as specified in 3.2.3.4.

4.8.15 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-STD-1190.

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4.9 Examinations. The material shall be visually examined to ensure conformance to the workmanship requirement specified in 3.5.

5. PACKAGING

5.1 Preservation, packing, and marking. Level C Packaging (see 6.2).

5.1.2 Level C. Preservation, packing, and marking shall be in accordance with MIL-STD-1190. Additional marking shall include, but not be limited to the following:

a. Lot or batch number.

b. Date of manufacture.

c. Type.

d. Width and yardage.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This material is intended for use as a compression molding material for high strength reinforced plastic components.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Type required (see 1.2 and 5.1.2).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1).

d. When first article is required (see 3.1 and 4.4).

e. First article sample size (see 4.4).

f. Quantity and roll size (see 5.1.2).

g. Width (see table I and 5.1.2).

h. Requirement for certification of storage life (see 3.2.5).

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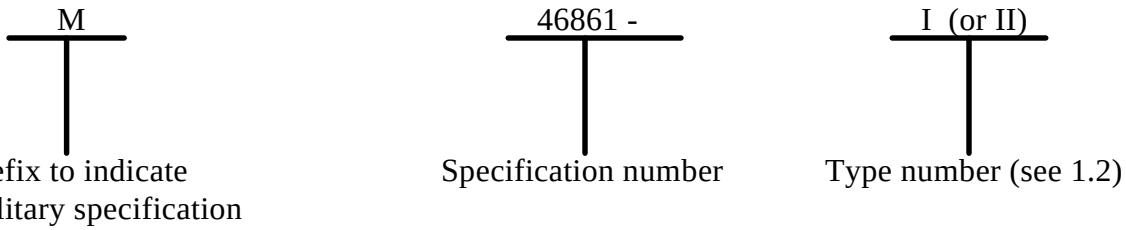
i. Requirements for preservation, packing and marking (see 4.8.15 and section 5).

j. Conditions for quality conformance inspection (see 4.5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Part or identifying number (PIN). The PIN to be used for molding material acquired to this specification is created as follows:



6.6 Subject term (keyword) listing.

Fabric, silicone
Preimpregnated fabric
Prepreg
Resin laminate
Silicon fiber

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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